



STEM Little Explorers

Summer of curiosity

30+ screen-free science adventures for long, sunny days,
sorted by age and ready for the backyard.

activities for ages 1-12





Welcome, summer explorers!

Long days, no homework, and a whole world to poke at. Inside are some of our favourite hands-on STEM activities — most use things you already have in the kitchen. Pick by your child's age, head outside, and let them get curious (and a little messy).

What's inside



Little explorers - sensory & squishy

ages 1-3 · toddlers



Curious preschoolers - kitchen magic

ages 3-5 · preschool



Junior scientists - backyard experiments

ages 6-8 · lower elementary



Master explorers - big-kid STEM

ages 9-12 · upper elementary



Summer survival kit for parents

for grown-ups



Summer science bingo



ages 1-3 · toddlers

Little explorers - sensory & squishy



Calm-down sensory bottle

⌚ 10 min Mess ● ● ●

Glitter swirling slowly through water - mesmerizing for little hands, and a sneaky lesson in density.

You'll need: a clear plastic bottle, water, clear glue, glitter

Full guide: stemlittleexplorers.com/en/homemade-sensory-bottle-child-sensory-development



Rainbow rice sensory bin

⌚ 20 min + drying Mess ● ● ●

Dye rice in summer colors for scooping, pouring and burying treasures. Great for the porch.

You'll need: dry rice, food colouring, zip-lock bags, a tray

Full guide: stemlittleexplorers.com/en/how-to-make-colored-rice



Squishy homemade playdough

⌚ 15 min Mess ● ● ●

Mix together your own homemade playdough: squish, roll and learn what makes it stretchy.

You'll need: flour, salt, water, oil, cream of tartar, food colouring

Full guide: stemlittleexplorers.com/en/make-homemade-playdough-learn-science



Color-mixing splash play

⌚ 15 min Mess ● ● ●

Two colors in, a brand-new one out: messy, hands-on color discovery for the water table.

You'll need: water, food colouring, clear cups, a tray

Full guide: stemlittleexplorers.com/en/make-sensory-play-colors



ages 3-5 · preschool

Curious preschoolers - kitchen magic



Magic colorful milk

⌚ 10 min Mess ● ● ●

A drop of soap sends colors exploding across the plate. Pure wow, zero cleanup stress.

You'll need: a plate, milk, food colouring, dish soap, a cotton bud

Full guide: stemlittleexplorers.com/en/make-colorful-milk-polarity-experiment



Groovy lava lamp

⌚ 10 min Mess ● ● ●

Bubbling blobs rise and fall in a bottle - a calm, hypnotic show for a hot afternoon.

You'll need: a clear bottle, water, vegetable oil, food colouring, an effervescent tablet

Full guide: stemlittleexplorers.com/en/make-lava-lamp



Erupting volcano

⌚ 15 min Mess ● ● ●

The classic fizzing eruption: best done outside where the foam can go everywhere.

You'll need: baking soda, vinegar, dish soap, food colouring, a small bottle

Full guide: stemlittleexplorers.com/en/how-to-make-homemade-volcano



Bouncy (glowing!) egg

⌚ 5 min + 2-day wait Mess ● ● ●

Soak an egg in vinegar for a few days and the shell vanishes, leaving a wobbly, bouncy egg.

You'll need: a raw egg, a glass, white vinegar

Full guide: stemlittleexplorers.com/en/egg-in-vinegar-make-bouncy-glowing-egg



ages 6-8 · lower elementary

Junior scientists - backyard experiments



Baking-soda rocket

⌚ 15 min Mess ● ● ●

Vinegar + baking soda = lift-off. A loud, messy, gloriously summer-outside launch.

You'll need: an empty plastic bottle, baking soda, vinegar, a cork, paper

Full guide: stemlittleexplorers.com/en/how-to-make-homemade-rocket-vinegar-baking-soda



Popsicle-stick catapult

⌚ 20 min Mess ● ● ●

Build it, then launch pom-poms across the lawn - a backyard physics tournament.

You'll need: craft sticks, rubber bands, a bottle cap, pom-poms

Full guide: stemlittleexplorers.com/en/make-popsicle-catapult



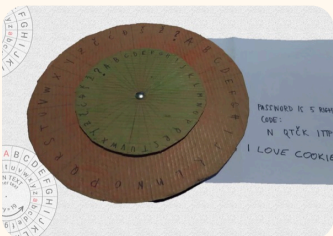
Grow gummy bears in water

⌚ 5 min + overnight Mess ● ● ●

Leave gummy bears soaking and watch them balloon overnight - a tasty intro to osmosis.

You'll need: gummy bears, water, glasses

Full guide: stemlittleexplorers.com/en/gummy-bears-osmosis-experiment



Secret cipher wheel

⌚ 20 min Mess ● ● ●

Make a decoder wheel and swap secret messages with summer-camp friends.

You'll need: card, a split-pin, scissors, a pen

Full guide: stemlittleexplorers.com/en/how-to-make-cipher-wheel



ages 9-12 · upper elementary

Master explorers - big-kid STEM



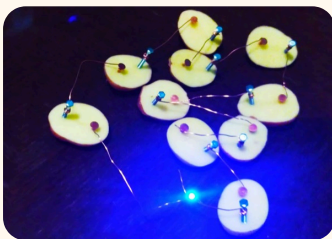
Backyard movie projector

⌚ 30 min Mess ● ● ●

Turn a shoebox + smartphone into a projector for an outdoor movie night under the stars.

You'll need: a shoebox, a magnifying glass, a smartphone, tape

Full guide: stemlittleexplorers.com/en/how-to-make-projector-using-smartphone



Potato battery

⌚ 20 min Mess ● ● ●

Power a small light from a potato - real electricity from the veggie drawer.

You'll need: a potato, copper wire, a galvanised nail, wires, an LED

Full guide: stemlittleexplorers.com/en/how-to-make-potato-battery



Write your name in binary

⌚ 25 min Mess ● ● ●

Decode how computers store letters, then string a binary-code name bracelet.

You'll need: string or pipe cleaners, pony beads in two colours

Full guide: stemlittleexplorers.com/en/write-your-name-in-binary



Grow rock candy crystals

⌚ 15 min + 1 week Mess ● ● ●

A week-long sugar-crystal experiment that ends in an edible, sparkly summer treat.

You'll need: sugar, water, a jar, string, a pencil

Full guide: stemlittleexplorers.com/en/make-sugar-crystals-stem-activity-kids



when the weather turns

Rainy-day rescue

Grey skies? These ones happily move indoors, to the kitchen table or the living-room floor.



Calm-down sensory bottle



Squishy homemade playdough



Magic colorful milk



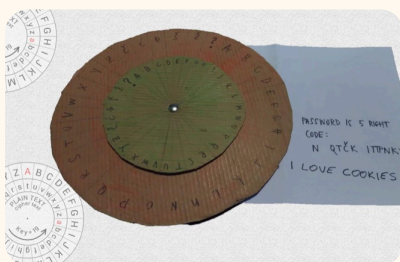
Groovy lava lamp



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Grow rock candy crystals



for grown-ups

Summer survival kit for parents



Beat “I’m bored” with a summer jar

Write each activity on a stick; when boredom strikes, they draw one. The choice is theirs, the calm is yours.



Embrace the mess - take it outside

Volcanoes, rockets and color splashes are a hose-down away from spotless when you run them in the yard.



Ask “what do you think will happen?”

One question turns a demo into real science. Let them guess, test and be wrong - that's the whole point.



Keep a summer discovery journal

A cheap notebook for drawings and “results” fights the summer slide and makes a keepsake by autumn.



Sun, water & safety first

Shade, hats and grown-up supervision near water and anything that launches, fizzes or pops.



Summer science bingo

Try one a day. Fill a row – or the whole card – for a proper summer of discovery!

 Calm-down sensory bottle 	 Rainbow rice sensory bin 	 Squishy homemade playdough 	 Color-mixing splash play 
 Magic colorful milk 	 Groovy lava lamp 	 Erupting volcano 	 Bouncy (glowing!) egg 
 Baking-soda rocket 	 Popsicle-stick catapult 	 Grow gummy bears in water 	 Secret cipher wheel 
 Backyard movie projector 	 Potato battery 	 Write your name in binary 	 Grow rock candy crystals 



Keep exploring all summer

This is just a taste. Find every experiment, plus new ones each week, on the site.

[Explore all activities ↗](#)

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